# EML 5131 | Combustion

Class Number: 25406

**Spring 2027**

## Instructor Information

**Name:** Claire Gregoire

## Course Details

**Catalog Description:** Chemical thermodynamics, chemical kinetics, flame propagation, detonation and explosion, combustion of droplets and spray.
**Pre- and Co-Requisites:** Prereq: EML 3100 or equivalent.
**Credit Hours:** 3
**Course Fees:** $0.00

## Required Materials

**AN INTRODUCTION TO COMBUSTION: CONCEPTS AND APPLICATIONS**
**ISBN:** 9781260575521
**Authors:** ALLAN T. KIRKPATRICK AND KENNETH K. KUO
**Publisher:** WILEY
**Edition:** 4TH

## A Note on Materials

The instructor has indicated that the third edition can also be used in this course.

## Course Goals and Objectives

At the end of this course, students should be able to:

1. **Core Combustion Principles:** Understand thermodynamics, reacting systems, energy conservation, and the First Law.

2. **Stoichiometry and Thermochemistry:** Calculate fuel–air ratios, theoretical and excess air, product composition, and heat of reaction.
3. **Chemical Equilibrium:** Analyze reaction direction and equilibrium compositions.
4. **Fuels and Energy Content:** Characterize fuel properties, heating values, and combustion performance
5. **Chemical Kinetics:** Analyze reaction rates and the effects of temperature, and reaction mechanisms.
6. **Combustion Emissions:** Understand pollutant formation and emissions reduction strategies.
7. **Deflagration and Detonation:** Explain propagation mechanisms, characteristics, and safety implications.
8. **Flame Dynamics and Propagation:** Analyze flame propagation, flammability limits, flame stability and structures, and fuel–oxidizer mixing.
9. **Computational Tools:** Use STANJAN to calculate equilibrium compositions and thermodynamic properties of reacting mixtures.

## Expectations and Student Learning Outcomes

***Relation to Program Outcomes (ABET):***

| Outcome | *Coverage |
|---|---|
| 1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics | High |
| 1. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors | Low |
| 1. An ability to communicate effectively with a range of audiences | Medium |
| 1. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts | Medium |

| Outcome | *Coverage |
|---|---|
| 1. An ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives | Low |
| 1. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions | Medium |
| 1. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies | High |

*Coverage is given as high, medium, or low. An empty box indicates that this outcome is not covered or assessed in the course.

## Methods of Evaluation

**(EML 5131):** Grades will be determined using these weights: 5% per Homework (25%), 15% for the Computational Combustion Project, and 20% per Exam (60%).

**(EML 4930):** Grades will be determined using these weights: 5% per Homework (25%), and 25% per Exam (75%). Co-listed undergraduate students will be tested as graduate students!

**Computational Combustion Project (EML 5131):** Project Description: Each project will be based on a problem from An Introduction to Combustion by Turns and Haworth. Whether projects are completed individually or in groups will depend on total course enrollment. The instructor will determine which problem is assigned to each student or group to ensure that different students or groups work on different problems.

- Students will solve the assigned problem using analytical and/or computational methods, present their methodology, explain their code, verify and interpret their results, and investigate how the results change when a relevant parameter or assumption is varied.

- Presentation Schedule: Presentations are intended to take place during the last week of class. Each presentation will be limited to 10 minutes, followed by 2–3 minutes for questions.

- Submission Requirements: Students must submit their presentation slides and complete source code, along with any necessary input files, to Canvas before their presentation. The submitted code must include detailed comments explaining the purpose and functionality of each line or section of code. Students must understand and be prepared to explain every line of code during the question period. A separate technical report is not required.

- Evaluation: Projects will be evaluated by the instructor and fellow class members according to the following criteria:

- Understanding of combustion theory and problem formulation (20%): Correct identification and application of governing equations, assumptions, and relevant combustion principles.
- Code implementation and computational methodology (20%): Correctness, organization, and clarity of the code, as well as the justification of the computational methods used.
- Results, physical interpretation, and verification (25%): Accuracy of results, appropriate verification of the solution, and interpretation of the physical significance of the findings.
- Additional investigation and analysis (15%): Investigation of how the results change when a relevant parameter or assumption is varied, with a clear explanation of the observed trends.
- Presentation quality and responses to questions (10%): Clarity, organization, effective communication of technical content, and ability to explain and defend the work.
- Code documentation and clarity (10%): Readability, detailed comments, and completeness of the submitted code and associated files.

- Peer Evaluation: Class members will contribute to the evaluation of presentations alongside the instructor. Students are expected to provide constructive feedback and assess presentations fairly according to the stated criteria.

## Grading Scale

Letter grade and percentage

| Letter | Percentage Value |
|---|---|
| A | 93 - 100% |
| A - | 90 - 92.9% |
| B + | 87 - 89.9% |
| B | 83 - 86.9% |
| B - | 80 - 82.9% |
| C + | 77 - 79.9% |
| C | 73 - 76.9% |
| C - | 70 - 72.9% |
| D + | 67 - 69.9% |
| D | 63 - 66.9% |
| D - | 60 - 62.9% |

| Letter | Percentage Value |
|---|---|
| E | 59.9% and below |

## Course Schedule

***Course Schedule**, Spring 2027*

| Week | Monday | Wednesday | Friday | Comments |
|---|---|---|---|---|
| 1 | 01/11 | 01/13 | 01/15 | Chapter 1 – Introduction. |
| 2 | ~~01/18~~ | 01/20 | 01/22 | Chapter 2 – Combustion and Thermochemistry. **01/18: Martin Luther King Jr. Day** |
| 3 | 01/25 | 01/27 | 01/29 | Chapter 2 – Combustion and Thermochemistry. |
| 4 | 02/01 | 02/03 | **02/05** | Chapter 2 – Combustion and Thermochemistry. **02/05: Homework 1.** |
| 5 | 02/08 | 02/10 | 02/12 | Chapter 6 – Coupling Chemical and Thermal Analyses of Reacting Systems. |
| 6 | 02/15 | **02/17** | **02/19** | **02/17: Midterm 1.** Practice Problems 02/15. **02/19: Homework 2.** |
| 7 | 02/22 | 02/24 | 02/26 | Chapter 17 – Fuels |
| 8 | 03/01 | 03/03 | **03/05** | Chapter 4 – Chemical Kinetics. **03/05: Homework 3.** |
| 9 | 03/08 | 03/10 | 03/12 | Chapter 4 – Chemical Kinetics. |
| 10 | **03/15** | **03/17** | **03/19** | **Spring Break** |
| 11 | 03/22 | **03/24** | **03/26** | **03/24: Midterm 2.** Practice Problems 03/22. **03/26: Homework 4.** |
| 12 | **03/29** | 03/31 | 04/02 | Chapter 15 – Emissions. **03/29: Seminar.** |
| 13 | 04/05 | 04/07 | **04/09** | Chapter 16 – Detonations. **04/09: Homework 5.** |
| 14 | **04/12** | 04/14 | 04/16 | Chapter 8 – Laminar Premixed Flames. **04/12: Seminar** (Tentative, to be confirmed). |
| 15 | **04/19** | **04/21** | **04/23** | **Project Presentations. Reading Days.** |
| 16 | 04/26 | 04/28 | **04/30** | **04/30: Final Exam, 7:30 – 9:30 am.** |

## University Policies and Resources

### UF Policies

Information about grading policies, support for students with disabilities, course evaluations, the Honor Code, and other course policies and campus resources can be found on the Syllabus Policies page.

### Responsible Use of Artificial Intelligence

See UF Policy 1-006 Responsible Use of Artificial Intelligence

#### Key Definitions

*Use*: This policy defines "use," in accordance with Policy 1-006: Responsible Use of Artificial Intelligence, as utilization of AI tools "in a manner that respects the rights and dignity of individuals and promotes fairness, transparency, and accountability." (1-006.4.1.4: Ethical, Responsible, and Accountable Use).

In addition, "use" is defined in accordance with Florida Board of Governors regulation 1.001 3o, "For purposes of this subsection, artificial intelligence functionality that is incidental to the ordinary operation of software or technology and does not generate, create, or materially modify content on behalf of the user is not considered use of artificial intelligence."

## Student Use of AI

**Some AI Use:**

The use of AI in graded assignments is permitted but limited as described with each assignment. Students should consult the directions for assigned work to determine what types of AI are allowed and how they can be used. The use of AI in graded assignments must be properly disclosed.

## Instructor Use of AI

**No AI Use:**

AI tools were not used to develop course materials and assessments, evaluate student work, or support course delivery.

## Instructor-Provided Statement

**AI Usage Policy for Computational Combustion Project (EML 5131):** The use of AI tools is permitted. However, students are fully responsible for the accuracy and validity of their work and must understand and be able to explain in detail every line of code submitted, including its purpose, functionality, and role in the overall solution. Students must disclose the use of AI tools and acknowledge external code or other substantial assistance.

## Instructor Use of AI to Assist Grading

**No AI:**

AI tools are not used to assist in grading

## Attendance Policy

### Excused and Unexcused Absences

Students may only participate in classes if they are registered officially or approved to audit with evidence of having paid audit fees. The Office of the University Registrar provides official class rolls to instructors.

Students are responsible for satisfying all academic objectives as defined by the instructor. Absences count from the first-class meeting.

Acceptable reasons for absence from or failure to engage in class include illness; Title IX-related situations; serious accidents or emergencies affecting the student, their roommates, or their family; special curricular requirements (e.g., judging trips, field trips, professional conferences); military obligation; severe weather conditions that prevent class participation; religious holidays; participation in official university activities (e.g., music performances, athletic competition, debate); and court-imposed legal obligations (e.g., jury duty or subpoena). Other reasons (e.g., a job interview or club activity) may be deemed acceptable if approved by the instructor.

For all planned absences, a student in a situation that allows an excused absence from a class, or any required class activity must inform the instructor as early as possible prior to the class. For all unplanned absences because of accidents or emergency situations, students should contact their instructor as soon as conditions permit.

Students shall be permitted a reasonable amount of time to make up the material or activities covered during absence from class or inability to engage in class activities because of the reasons outlined above.

If a student does not participate in at least one of the first two class meetings of a course or laboratory in which they are registered, and they have not contacted the department to indicate their intent, the student can be dropped from the course. Students must not assume that they will be dropped, however. The department will notify students if they have been dropped from a course or laboratory.

The university recognizes the right of the instructor to make attendance mandatory and require documentation for absences (except for religious holidays), missed work, or inability to fully engage in class. After due warning, an instructor can prohibit further attendance and subsequently assign a failing grade for excessive absences.

## Religious Holidays Guidelines

At the University of Florida, students and faculty work together to allow students the opportunity to observe the holy days of their faith. A student should inform the faculty member of the religious observances of their faith that will conflict with class attendance, with tests or examinations, or with other class activities prior to the class or occurrence of that test or activity. The faculty member is then obligated to accommodate that particular student's religious observances. Because students represent a myriad of cultures and many faiths, the University of Florida is not able to assure that scheduled academic activities do not conflict with the holy days of all religious groups. Accordingly, individual students should make their need for an excused absence known in advance of the scheduled activities.

The Florida Board of Education and state law govern university policy regarding observance of religious holidays.

**Guidelines**

- Students, upon prior notification to their instructors, shall be excused from class or other scheduled academic activity to observe a religious holy day of their faith.
- Students shall be permitted a reasonable amount of time to make up the material or activities covered in their absence.
- Students shall not be penalized due to absence from class or other scheduled academic activity because of religious observances.

If a faculty member is informed of or is aware that a significant number of students are likely to be absent from class because of a religious observance, the faculty member should not schedule a major exam or other academic event at that time.

A student who is to be excused from class for a religious observance is not required to provide a second party certification of the reason for the absence. Furthermore, a student who believes that they have been unreasonably denied an education benefit due to religious beliefs or practices may seek redress through the student grievance procedure.

## Absence due to Illness

A student who is absent from class or any required class-related activity because of illness should contact their instructor, if feasible, as early as possible prior to the missed class or activity.

Students shall be permitted a reasonable amount of time to make up the material or activities covered during an excused absence.

Students should contact their college by the deadline to drop a course for medical reasons. Students can petition the Dean of Students Office to drop a course for medical reasons. The university's policy regarding medical excuse from classes is maintained by the Student Health Care Center.

## Twelve-Day Rule

Students who participate in university-sponsored athletic or scholarly activities are permitted to be absent 12 scholastic days per semester without penalty. A scholastic day is any day on which regular class work is scheduled as defined in the approved university calendar. *For information on academic dates and deadlines, visit the UF Academic Dates and Deadlines webpage.*

The student or student's advisor must notify the instructor as early as possible prior to the anticipated absence to allow ample time for accommodations. Instructors must be flexible and not penalize students when re-scheduling during-term and final exams, class assignments, and other required activities and must follow the UF Attendance Policy herein and UF Examination Policies. As noted in the UF Examination Policies, during-term exams should be re-scheduled no later than before the end of the semester, while final exams no later than 90 days after the originally scheduled exam time. However, instructors are encouraged to re-schedule final and during-term exams, assignments, and other activities as soon as possible after the last day of the absence and must not penalize the student in any way. *For additional information, refer to the UF Examination Policies and Reading Days webpage.*

A group's schedule that requires absence of more than 12 scholastic days should be adjusted so that no student is absent from campus more than 12 scholastic days. Students who previously have been warned in writing by their instructor about the impact of absences on their individual class performance should not incur additional absences, even if they have not been absent 12 scholastic days. The student is responsible to maintain satisfactory academic performance and attendance.

## Course Policies and Resources

**Policies and Procedures:** Since this course has both undergraduate (EAS 4930) and graduate (EML 5131), in order to distinguish between the student's levels, students registered for EML 5131 will be required to perform a Computational Combustion Project and periodically an extra problem in HW assignments.

**Attendance Policy, Class Expectations, and Make-Up Policy:** Attendance at all lectures is expected and will occasionally be monitored. Requirements for class attendance and make-up exams, assignments, and other work in this course are consistent with university policies. Click here to read the university attendance policies: https://catalog.ufl.edu/UGRD/academic-regulations/attendance-policies/

**Homework:** Five homework assignments will be given during the semester, with each assignment provided two weeks before its due date. Periodically, problems will be assigned with a specified due date. All homework assignments must be submitted through the e-learning website. Late homework will not be accepted. Hardship cases will be considered on an individual basis and only when the instructor has been contacted before the assignment due date.

**Exams:** Two exams will be administered during the semester at the regularly scheduled class times. Excused absences will be granted only for valid reasons consistent with university policy. Students who anticipate a conflict or have a valid reason for missing an exam must contact the instructor before the scheduled exam time. Alternative arrangements may be considered when justified by the circumstances and consistent with university policy. The final exam date and time have been scheduled by the University: 04/30, 7:30–9:30 a.m.

**Grade Questions and Regrade Requests:** Students are expected to review their grades and feedback promptly. Any questions or requests for reconsideration of a grade must be submitted to the instructor within one week of the grade being posted. Requests submitted after this period may not be considered.

## Classroom Behavior

**In-Class Recording:** Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are (1) for personal educational use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil

proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor.

Publication without permission of the instructor is prohibited. To “publish” means to share, transmit, circulate, distribute, or provide access to a recording, regardless of format or medium, to another person (or persons), including but not limited to another student within the same class section. Additionally, a recording, or transcript of a recording, is considered published if it is posted on or uploaded to, in whole or in part, any media platform, including but not limited to social media, book, magazine, newspaper, leaflet, or third party note/tutoring services. A student who publishes a recording without written consent may be subject to a civil cause of action instituted by a person injured by the publication and/or discipline under UF Regulation 4.040 Student Honor Code and Student Conduct Code.